MIL-S-23492B
3 August 1971
SUPERSEDING
MIL-S-23492A(SHIPS)
2 February 1970
(See 6.5)

MILITARY SPECIFICATION

SCALER, NEEDLE TYPE, PNEUMATIC, PORTABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable, pneumatic, needle type scalers suitable for use in the removal of paint, rust, and scale from flat and irregular surfaces.

1.2 Classification. Scalers covered by this specification shall be of the following sizes and styles, as specified (see 6.2):

Size 1 - Scaler, 6 pounds, 19 inches.
Style B - Inline handle
Style C - Pistol grip handle

Size 2 - Scaler, 8.5 pounds, 20 inches.
Style A - Offset handle
Style B - Inline handle
Style C - Pistol grip handle
Style D - Open handle

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
GGG-C-330 - Chisel, Tool Blank, Calking Tool and Metal-Forming Tool, Power Hammer.
PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY
MIL-S-20166 - Steel Structural Shapes, Weldable Medium Carbon and High Tensile; Hull and Structural.

STANDARD

FEDERAL
FED-STD-66 - Steel, Chemical Composition and Hardenability.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

FSC 5130

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
S5.1-1971 - Measurement of Sound from Pneumatic Equipment, Test Code for the.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

DEPARTMENT OF LABOR
PART II - Occupational Safety and Health Standards (OSHA)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.)

3. REQUIREMENTS

3.1 Qualification. Portable, pneumatic scalers furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Illustrations. The illustrations shown herein are descriptive, not restrictive, and are not intended to preclude the purchase of scalers otherwise conforming to this specification.

3.3 General. The scaler shall impact a bundle of steel needles against the surface to be cleaned in a manner that will permit individual needles to impact at various planes, thus covering all points of an irregular surface within the periphery of the needle bundle. The scaler shall consist of hardened needles, an actuating mechanism to drive the needles, a housing, throttle, air connection fitting, and a handle.

3.4 Materials. The materials used in the construction of items and components shall be sound, of uniform quality and condition, and shall conform in composition, heat treatment, and suitability to the standard practices of manufacturers producing tools of the sizes and styles required in this specification. All items and parts shall be new and high grade commercial quality.

3.5 Construction. Scalers shall be of heavy duty construction and designed for continuous duty in an industrial ship repair environment. Scalers shall be reliable and effective when operated with compressed air between 80 and 100 pounds per square inch gage (psig). Scalers shall be self-contained and adjustable, and parts shall be accessible for replacement and repair without requiring special tools, other than tools normal to the trade. Parts which require lubrication shall be properly enclosed to prevent entrance of foreign particles and leakage of lubricant. Working parts subject to wear shall be heat treated to withstand the required service (see table I). Parts dimensions and configuration shall be standardized so that similar parts may be exchanged between scalers of the same size and style without degradation of the tools effectiveness.

3.6 Dimensional and performance characteristics. Dimensional and performance characteristics of the scalers shall conform to the requirements of table I.

Table I - Dimensional and performance characteristics.

	Size 1	Size 2
Weight, with needles (max)	6 pounds	9.5 pounds
Length, overall, with needles (max)	19 inches	20 inches
Hardness of needles (Rockwell C)	52 - 60	52 - 60
Hardness of piston (Rockwell C)	54 - 60	54 - 60
Air consumption (free air per min) at 90 psig (max)	14 SCFM	13 SCFM
Blows per minute at no load (min)	2000	2000
Impact power (minimum metal removing capability when tested in accordance with 4.6.7)	5 grams/min	10 grams/min

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3.7 Air inlet connection. The air inlet connection shall consist of a steel fitting threaded with internal American Standard taper pipe threads (National Pipe Threads (NPT) or National Pipe Threads Fine (NPTF)) conforming to Handbook H-28.

3.8 Strainer. Scalers shall be fitted with an air strainer or screen which shall be effective in retaining solid particles in the compressed air supply. The strainer shall be of such design that it can be readily removed for cleaning.

3.9 Identification marking. Scalers shall be permanently marked with the manufacturer's style and size designation, together with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.10 Handles. The handle shall be located opposite the needle end of the scaler and shall be style A, B, C, or D as illustrated in figures 1, 2, 3, and 4, respectively, and as specified (see 6.2).

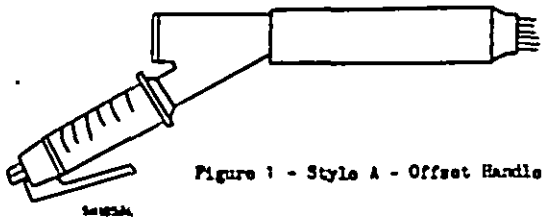


Figure 1 - Style A - Offset Handle.

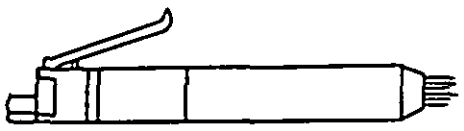


Figure 2 - Style B - Inline Handle.

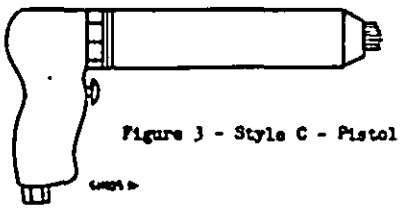


Figure 3 - Style C - Pistol Grip Handle.

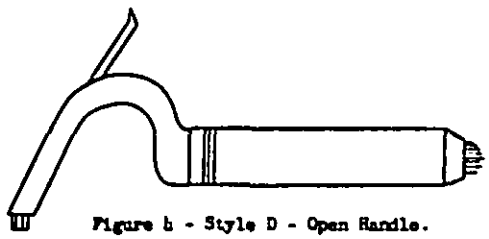


Figure 4 - Style D - Open Handle.

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3.11 Throttle. The throttle shall be manually operated and shall automatically and positively shut off the air supply to the tool when released. The operating trigger shall be located on the handle and shall be of a type to insure ease of operation under service conditions. The valve trigger shall be fitted with "O" rings or gaskets and prevent air leakage in either the open or closed position of the throttle valve.

3.12 Needles. A minimum of 18 needles shall be provided. Needles shall have a minimum length of 6 inches, shall be installed parallel to housing, and shall extend 2-1/2 inches beyond the nose end of the scaler. The needle hardness shall conform to table I. The needle body diameter shall be 0.120 ± .005 inch, the needle head diameter shall be 0.185 ± 0.005 inch and the needle head height shall be 3/32 ± 1/32 inch.

3.13 Needle holder. The diameter of the needle hole in the needle holder shall be 0.135 ± 0.008 inch.

3.14 Housing. The housing shall be of forged steel, tubing, casting, or alloy, heat treated and of thickness that will prevent distortion, breakage, or excessive wear due to impact under normal tool operation. It shall be designed to house the operating mechanism, the scaling needles, and to retain needle holder and to prevent it from becoming a projectile. The needle end of the housing shall be either round or rectangular as specified in 6.2.

3.15 Depth of irregular surface penetration. When tested as specified in 4.6.6, the needles shall make contact with the backing plate of the weld joint assembly.

3.16 Endurance test. After completion of the endurance test specified in 4.5, the scaler shall have no broken, cracked, or deformed parts, other than normal wear of the needles. Failure of the scaler to comply with endurance test shall be cause for rejection.

3.17 Chipping test. When tested as specified in 4.6.7, the scalers shall remove the minimum amount of metal specified in table I.

3.18 Service test. After completion of the service test as specified in 4.7, the scaler shall have no broken, cracked, or deformed parts, other than that acquired by normal wear of the needles.

3.19 Safety requirements. Needle scalers shall comply with the applicable provision of OSHA as related to portable power tools and 3.19.1.

3.19.1 Sound pressure levels. The sound pressure levels measured at an unobstructed distance of 1 meter, shall not exceed 90 decibels on A-weighted scale (dBA) on the weighted scale in any prescribed location (see 4.3).

3.20 Workmanship. The workmanship shall be free from all imperfections which may affect the serviceability of the tool.

3.21 Instruction book. The manufacturer shall furnish two copies of his regular instruction book and parts list with each needle scaler. The instruction book shall include all illustrations, instructions, and other necessary data for safety operation, maintenance, adjustment, and repair of the tool. The parts list shall include views of parts identified by manufacturer's part number, and ordering instructions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the examinations of 4.4 and the tests of 4.5, 4.6, 4.7, and 4.8. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3 and 6.3.1).

4.2.1 The manufacturer shall submit for testing, one scaler complete with instruction book parts lists, and one extra set of needles for each size and style scaler for which qualification tests have been authorized.

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4.3 Sampling for quality conformance inspection.

4.3.1 Lot. Scalers of the same size presented at one time shall be considered a lot for the purpose of quality conformance inspection.

4.3.2 Sampling for examination. A random sample of scalers shall be selected from each lot for examination of visual and dimensional characteristics in accordance with MIL-STD-105 at inspection level III for lots of 40 and under, inspection level II for lots of 41 to 300 and inspection level I for lots of 301 and over. The acceptable quality level shall be 1.5 percent defective.

4.3.3 Sampling for tests. A random sample of scalers shall be selected from each lot for the examination of 4.4 and tests of 4.6, 4.6.1, 4.6.2, 4.6.3, 4.6.4, and 4.6.5, in accordance with MIL-STD-105 at inspection level S-4. The acceptable quality level shall be 1.5 percent defective.

4.4 Examination. Each scaler offered shall be visually examined, dimensionally measured, statically tested, and weighed to verify conformance with the following requirements:

(a) Maximum weight and size (table I).
(b) Needle dimensions and hardness (table I and 3.12).
(c) Needle holder hole dimension (3.13).
(d) Hardness of needles and piston (table I).
(e) Inclusion of all specified components including air connection fitting and strainer.

4.5 Endurance test. Upon satisfactory completion of the visual and dimensional examination of 4.4, and before any other test is conducted, an endurance test shall be performed as follows to determine conformance with 3.16.

4.5.1 Apparatus. The test apparatus shall be in accordance with figure 5 and shall consist of the following:

A steel base plate 1-3/16 inch thick, 4-1/2 inches wide and 34 inches long. Two split holding blocks 1/2 by 2-3/4 by 4 inches are attached to the base and are used to hold the needle scaler. A locking device for locking the needle scaler in place is also attached to the base plate. A striking plate made of 4 inch angle plate is attached to the base plate as is the striking plate locking device. A test specimen, against which the needles will impact, is backed by rubber and attached to the striking plate. The test specimen is 3/8 inch thick steel, 4 by 4 inches and conforming to MIL-S-20166. The rubber backing is 1/2 inch thick and also 4 by 4 inches and has a Shore "A" Durometer hardness of 60 to 70.

4.5.2 Procedure. The scaler shall be positioned in the split holding blocks with the needles fully retracted. Using an appropriate spacer, the needles shall be positioned 1/32 inch from the test plate and the scaler locked in position. The scaler shall be operated continuously for not less than 3 hours at 90 psig with the needles impacting against the test plate. Upon completion of the endurance test, the tool shall be disassembled and the various parts examined for conformance with the requirements of 3.16.

4.6 Performance tests.

4.6.1 Performance tests conditions. All performance tests shall be based on operation of the scalers with compressed air with a gage pressure of 90 psig measured at the end of the hose attached to the air meter. Test scalers shall be connected to the air supply by a 10-foot length of 5/8-inch inside diameter hose connected to the tool with a coupling conforming to table II.

Table II - Coupling requirements.

Air-inlet connection of tool (iron pipe size)	Inside diameter of hose coupling
Inch	Inch
1/4	0.250
3/8	.375
1/2	.500
3/4	.500

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4.6.2 Air consumption. Scalers shall be subjected to a test for air consumption to determine compliance with table I.

4.6.3 Blows per minute. Scalers shall be subjected to a test for blows per minute at no load to determine compliance with table I.

4.6.4 Air leakage. Throttle shall be tested for tightness. Scaler shall be pressurized to 150 psig with throttle closed for a period of 15 minutes. No pressure drop shall be permitted at the end of 15 minutes.

4.6.5 Retainer test. Scaler shall be connected to a rated power source, and operated "free in air" (other than under load) for a 20 second duration. Any separation of the driver or needle holder from the tool shall constitute failure of this test.

4.6.6 Irregular surface depth penetration test. The irregular surface depth penetration test shall be performed as follows to determine conformance with 3.15.

4.6.6.1 Test apparatus. Test apparatus shall be the same equipment specified in 4.5.1.

4.6.6.2 Irregular surface test plate. Irregular surface test plate shall be a single "V" weld joint assembly which is attached to the striking plate of the test apparatus. The test plate shall be made of 0.375-inch thick plate, double beveled 30 degree angles with a 0.25 inch root opening at the backing plate. A piece of stiff cardboard shall be placed between the backing plate and the beveled plates. The test assembly shall be held together with screws.

4.6.6.3 Procedure. The test scaler shall be placed in the split holding blocks. The needles shall be placed in the retracted position. The scaler shall then be positioned so that the needles will be 1/32 inch from the test specimen. This shall be accomplished by the use of a spacer. The scaler shall then be locked in place and operated for one minute. After completion of the test the cardboard shall be examined for indications that the needles have made contact with the backing plate as required by 3.15.

4.6.7 Chipping test. The chipping test shall be performed as follows to determine conformance with 3.17.

4.6.7.1 Apparatus. The test apparatus shall consist of a chipping hammer testing machine developed to test portable, pneumatic chipping hammers. The salient features in the design of the testing machine are as follows:

4.6.7.1.1 Frame. The machine consists of two upright channels supporting two horizontal angles on which are fastened two machined tracks. The assembly is mounted on a heavy base plate.

4.6.7.1.2 Method of holding plate. The plate to be used for evaluating the needle scalers is held rigidly in position in the frame between the upright channels by means of bolts bearing against side-plates which, in turn, bear against the plate. A vertical edge of the plate is firmly butted against the web of the upright channel at the front of the machine by screwing two bolts tightly against the opposite vertical edge of the plate.

4.6.7.1.3 Mounting of needle scaler. The needle scaler is mounted in a carriage. The carriage holds the scaler inclined at an angle of 15 degrees to the edge of the plate being chipped. The vertical channel at the front of the machine is fitted with pulleys carrying a wire rope that is attached to the carriage and weighted at the other end. The wire rope pulls in a line that passes approximately through the center of gravity of the loaded carriage. A light oil is used as a lubricant between the carriage and the tracks. Resistance to static and sliding friction is equal to 15 pounds and 10 pounds, respectively.

4.6.7.1.4 Tracks and cutting arrangement. Chipping is done by a cutter that moves in grooves machined into the inside edges of the tracks. The squared end of a pneumatic chisel blank impacts on an intermediate piece which moves the cutter forward. This intermediate piece rides in the same grooves as the cutter, and is so shaped that the center axis of the chisel passes through the center of gravity of the intermediate piece.

4.6.7.1.5 Control of thickness of chip. The desired thickness of metal to be chipped is obtained by a gear train, calibrated in thousandths of an inch, which raises and lowers the base to which the plate is attached.

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4.6.7.2 Test conditions for determining the performance of needle scalers are as follows:

4.6.7.2.1 Cutters.

(a) Shall be 2.5 inches long by 2 inches wide by 5/16 inch thick, and shall be used until they have been reduced to a length of not less than 2 inches.
(b) Shall be forged from chisel stock meeting the requirements of class D material of GGG-C-330.
(c) Shall be heat treated to a hardness of Rockwell "C" 55 to 60.

4.6.7.2.2 The angle of the cutter shall be 57.5 degrees, and shall be closely maintained by frequent honing and grinding.

4.6.7.2.3 The clearance between the cutter and the grooves in the tracks shall be maintained between 0.002 and 0.004 inch.

4.6.7.2.4 A clearance shall be provided on the underside and to the rear of the cutting edge and shall taper backward 0.006 inch per inch of length of the cutter.

4.6.7.2.5 Steel test plates.

(a) The plates shall have a Brinell hardness of 110 to 120 and shall conform to the requirements for carbon steel plate, type 1020 of FED-STD-66.
(b) Shall be 3/16-inch thick, 20 inches wide, and initially 30 inches long. Not more than 1/2 of the length of the plate shall be consumed in tests.
(c) Chipping shall be in the direction of rolling.

4.6.7.3 Procedure. The needles and spring of the needle scaler shall be replaced with a chisel blank 7.5 inches long with a squared end. The modified scaler shall then be locked into the carriage of the testing machine at an angle of 15 degrees to the horizontal. Size 1 scalers shall be pulled by a 25 pound load applied in a horizontal direction on the carriage. Size 2 scalers shall be pulled by a 50 pound load. The scaler shall be operated with the throttle fully open. The dimensions of the cut shall be: width - 3/16 inch, length - 20 inches, and depth - 0.050 ± 0.005 inch. The time required to make a 20-inch long cut shall be recorded. The chip removed shall be weighed and the number of grams of metal removed per minute calculated. This test shall be conducted three times and the average of the results of the tests shall be used to determine conformance to the requirements of 3.17, and table I.

4.7 Service test. Upon completion of the examinations in 4.4, endurance test in 4.5, and the performance tests in 4.6, the scaler shall satisfactorily complete 100 hours of actual work under practical working conditions. The tests shall determine the ease of operation, general serviceability, and durability of the tool. Failure of any component of the tool shall be cause for rejection.

4.7.1 Examination after service test. Upon completion of the service test, the tool shall be disassembled and the various parts examined. Any tool having broken, cracked, or deformed parts, other than normal wear of the needles, shall be rejected. Tools satisfactorily completing the service test shall again be subjected to the chipping test of 4.6.7 and shall conform to the applicable requirements of 3.18.

4.8 Sound pressure level test. The sound pressure level test shall be conducted in accordance with the applicable provisions of ANSI S5.1-1971 except as modified herein. The acceptable limits shall be as specified in 3.19.1.

4.8.1 Acoustical calibration. The entire instrumentation system shall be calibrated before each test series allowing a minimum instrument warm-up period of 5 minutes if transistorized and 1/2 hour if vacuum tube. Recalibration shall be performed if any instrument adjustment is necessary during progress of the test.

4.8.2 Background sound level. The ambient noise level shall be at least 10 decibels (dB) less than the measurement taken with the tool running in all tool and microphone orientations.

4.9 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing, and marking shall be in accordance with the requirements of section 5 and the document specified therein.

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5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation, packaging, packing, and marking. Scalers shall be preserved, packaged, packed, and marked in accordance with PPP-P-40. The level of preservation and packaging, and the level of packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. Portable pneumatic scalers covered by this specification are intended for use in the removal of paint and scale from flat and irregular metal surfaces.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of scaler required (see 1.2 and table I).
(c) Style of scaler required (see 3.10 and figures 1, 2, 3, and 4).
(d) Shape of scaler required (see 3.14).
(e) Level of preservation and packaging, and packing required (see 5.1).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-23492 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 THE MARGINS OF THIS SPECIFICATION ARE MARKED "*" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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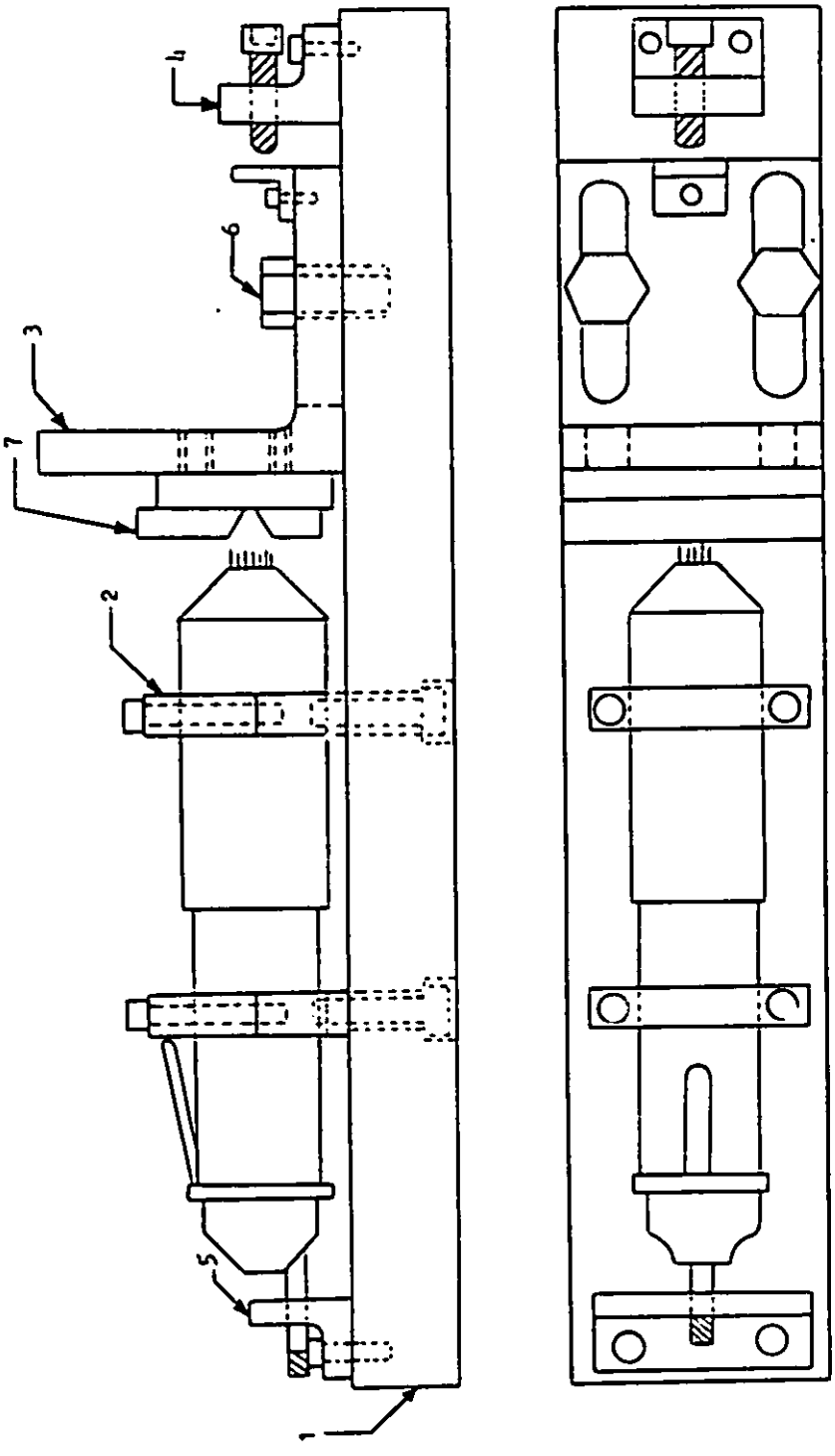


Figure 5 - Endurance test fixture with irregular surface penetration test specimen in place.

1. Base plate 1-3/16 by 4-1/2 by 34 inches.
2. Split holding blocks 1/2 by 2-3/4 by 4 inches.
3. Striking plate 4-inch angle plate.
4. Striking plate locking device.
5. Needle scaler locking device.
6. 3/4-inch - 10 bolts.
7. Test specimen assembly.

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